FINDING ROOTS

Factoring $x^3 - 4x$ completely

Two moves turn the cubic into a product, and each factor names one of its three roots: 0, 2 and -2.

01

EXPANDED — BEFORE

$$x^3 - 4x$$

↓ Extract the common factor — take out x

02

COMMON FACTOR OUT

$$x(x^2 - 4)$$

↓ Difference of squares — $x^2 - 4 = (x - 2)(x + 2)$

03

FULLY FACTORED — AFTER

$$x(x - 2)(x + 2)$$